

Úloha 1.1

Upravujme pravou stranu rovnice:

$$\begin{aligned}
 & \frac{1}{4} (\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2) = \frac{1}{4} (\langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle - \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle) \\
 &= \frac{1}{4} ((\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle) - (\langle \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle)) \\
 &= \frac{1}{4} (2 \langle \mathbf{u}, \mathbf{v} \rangle + 2 \langle \mathbf{v}, \mathbf{u} \rangle) = \frac{1}{2} (\langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle}) \\
 &= \frac{1}{2} \cdot 2 \operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) = \operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle),
 \end{aligned}$$

neboť $x + \bar{x} = 2 \operatorname{Re}(x)$.

Obdobně dostaneme i druhou identitu:

$$\begin{aligned}
 & \frac{1}{4} (\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2) = \frac{1}{4} (\langle \mathbf{u} - i\mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle - \langle \mathbf{u} + i\mathbf{v}, \mathbf{u} + i\mathbf{v} \rangle) \\
 &= \frac{1}{4} ((\langle \mathbf{u}, \mathbf{u} \rangle + (-i) \langle \mathbf{u}, \mathbf{v} \rangle + \overline{(-i)} \langle \mathbf{v}, \mathbf{u} \rangle + \overline{(-i)} \cdot (-i) \langle \mathbf{v}, \mathbf{v} \rangle) \\
 &\quad - (\langle \mathbf{u}, \mathbf{u} \rangle + i \langle \mathbf{u}, \mathbf{v} \rangle + \bar{i} \langle \mathbf{v}, \mathbf{u} \rangle + \bar{i} \cdot i \langle \mathbf{v}, \mathbf{v} \rangle)) \\
 &= \frac{1}{4} (-i \langle \mathbf{u}, \mathbf{v} \rangle + i \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle - i \langle \mathbf{u}, \mathbf{v} \rangle + i \langle \mathbf{v}, \mathbf{u} \rangle - \langle \mathbf{v}, \mathbf{v} \rangle) \\
 &= \frac{1}{4} (-2i \langle \mathbf{u}, \mathbf{v} \rangle + 2i \langle \mathbf{v}, \mathbf{u} \rangle) = \frac{1}{2} \cdot i \cdot (-\langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle}) \\
 &= \frac{1}{2} \cdot i \cdot (-2i) \operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) = \operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle),
 \end{aligned}$$

neboť pro $x = a + bi$ je $-x + \bar{x} = -(a + bi) + (a - bi) = -2bi = -2i \operatorname{Im}(x)$. □